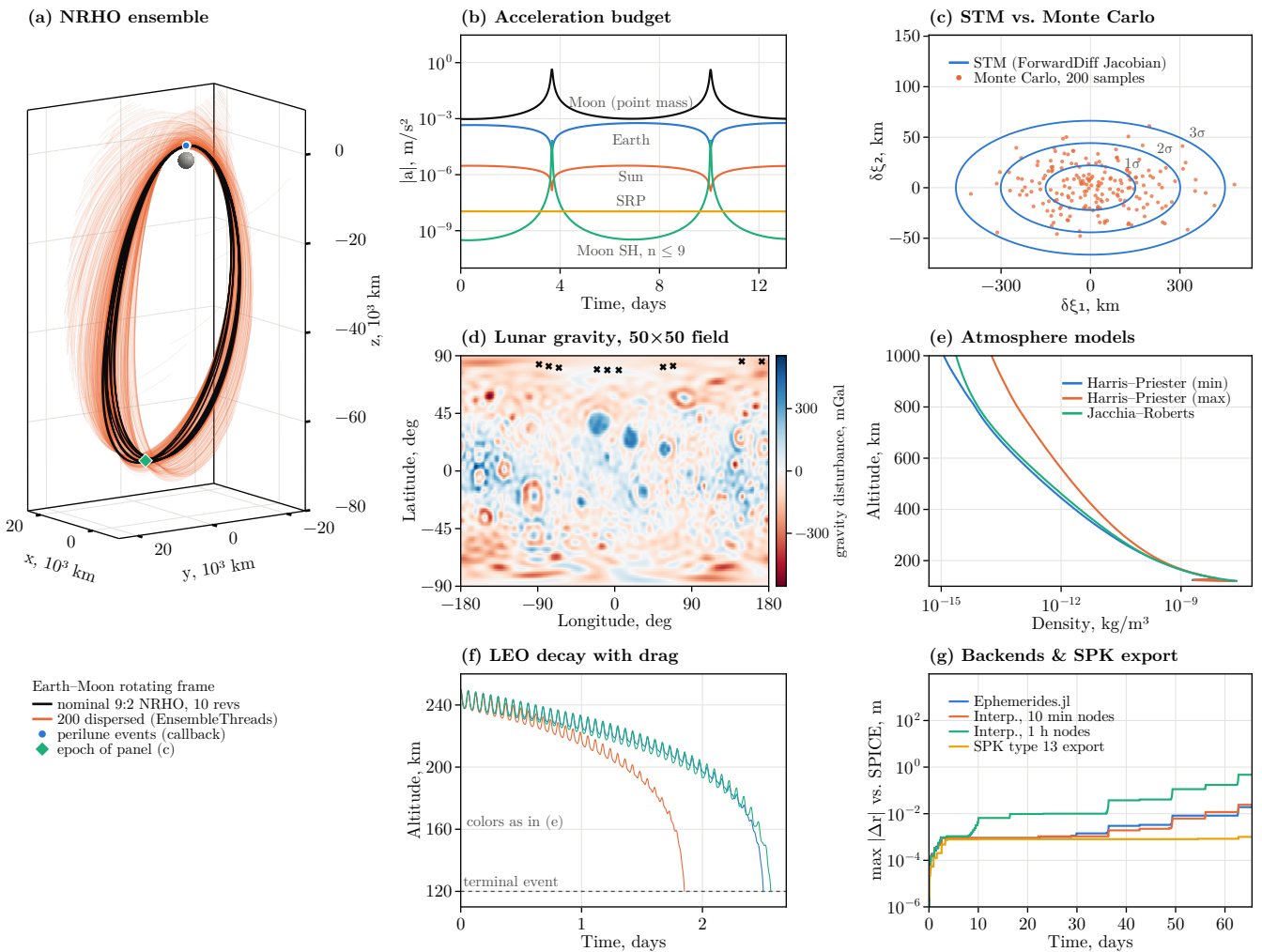

(a) NRHO ensemble
x, 10³ km
y, 10³ km
z, 10³ km
Earth–Moon rotating frame
nominal 9:2 NRHO, 10 revs
200 dispersed (EnsembleThreads)
perilune events (callback)
epoch of panel (c)
(b) Acceleration budget
Moon (point mass)
Earth
Sun
SRP
Moon SH, n ≤ 9
|a|, m/s²
Time, days
(c) STM vs. Monte Carlo
STM (ForwardDiff Jacobian)
Monte Carlo, 200 samples
1σ
2σ
3σ
δξ₂, km
δξ₁, km
(d) Lunar gravity, 50×50 field
Latitude, deg
Longitude, deg
gravity disturbance, mGal
(e) Atmosphere models
Harris–Priester (min)
Harris–Priester (max)
Jacchia–Roberts
Altitude, km
Density, kg/m³
(f) LEO decay with drag
colors as in (e)
terminal event
Altitude, km
Time, days
(g) Backends & SPK export
Ephemerides.jl
Interp., 10 min nodes
Interp., 1 h nodes
SPK type 13 export
max |Δr| vs. SPICE, m
Time, days